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B. E. (Sixth Semester) Examination, April-May 2020

(New Scheme)

(Mechanical Engg. Branch)

POWER PLANT ENGINEERING

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Attempt all questions. Part (a) from each question is compulsory. Attempt any two parts from parts (b), (c) and (d) of each question. Use of steam table & Mollier's chart is permitted.*

Unit-I

1. (a) Name the various sources of Energy. 2

(b) Explain the primary and secondary electrical distribution system with neat diagram. 7

(c) Explain the modern layout of Steam Power Plant with diagram. 7

(d) Explain the site selection of Diesel Power Plant. 7

Unit-II

2. (a) Draw T-s and h-s diagram of reheat cycle. 2

(b) Explain the various stages of Coal Handling Plant. 7

(c) Explain the thermal feed water treatment. 7

(d) A turbine applied with steam at a pressure of 32 bar, 410°C. The steam then expands isentropically to a pressure of 0·08 bar. Find the dryness fraction at the end of expansion and thermal efficiency of the cycle.

If the steam is reheated to 5·5 bar to a temperature of 400°C and then expands isentropically to a pressure of 0·08 bar. What will be the dryness fraction and thermal efficiency of the cycle? 7

Unit-III

3. (a) Define Hydro Electric Power Plant. 2

(b) Compare Steam Power Plant, Hydro Power Plant and Diesel Power plant. 7

(c) The following data is available from the investigation of a hydro power site. Available head = 50 m, Total catchment area = 50 sqkm, Rainfall = 1·5 m per year, Only 70% of the rainfall can be utilized.

Turbine efficiency = 90%, Penstock efficiency = 75%, Generator efficiency = 95%.

Calculate the actual power and also the suitable capacity of a generator. 7

(d) The following are the observation given for a single cylinder oil engine on dual cycle.

Oil consumption = 10·2 kg/hr

Calorific value of fuel = 43,890 kJ/kg

Air consumption = 3·8 kg/min

Speed = 1900 rpm

Torque = 186 Nm

Quantity of cooling water = 15·5 kg/min

Temperature rise = 36°C

Exhaust Gas Temperature = 410°C

Room Temperature = 20°C

Cp for Exhaust gas = 1·17 kJ/kgK

Calculate : 7

(i) Brake power

(ii) Brake specific fuel consumption

(iii) Brake thermal efficiency

(iv) Draw heat balance sheet on minute basis.

Unit-IV

4. (a) Define chain reaction. 2

(b) Explain CANDU reactor with neat diagram. 7

(c) Write the difference between nuclear fission & nuclear fusion. 7

(d) A nuclear reactor consumes 10 kg of Uranium per day. Calculate its power output if the average energy released per U^{235} fission is 200 MeV. 7

Unit-V

5. (a) Define Load Factor. 2

(b) The maximum demand of power station is 96 MW & installed capacity is 110 MW. The daily load curve is describe as below :

Time in hrs	0-6	6-8	8-12	12-14	14-18	18-22	22-24
Load in MW	48	60	72	60	84	96	48

Determine : 7

(i) Draw load curve & load duration curve

(ii) Load factor

(iii) Plant use factor

(c) For a 200 MW thermal power plant the following are the data :

Fixed cost = Rs 24×10^6 for a year

Caloric value of fuel = 20,000 kJ/kg

Other expenses (like salary & consumable) Rs 280 /kW per year

Plant heat rate = 18000 kJ/K Whr at 100% capacity factor. Determine generating cost per K Whr at 100% capacity factor. 7

(d) Define the following : 7

(i) Maximum demand

(ii) Utilisation factor

(iii) Plant Capacity Factor

(iv) Plant Use factor

(v) Demand Factor

(vi) Diversity Factor

(vii) Connected load